

TOI-1259Ab

R-2053

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1259_230829 · created 2026-08-09T00:35:47+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460185.7117 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.201 +/- 0.06
Transit depth (Rp/Rs)^2	4.06 +/- 2.42 [%]
Orbital Inclination (inc)	87.31 +/- 1.47
Airmass coefficient 1 (a1)	2.1 +/- 0.17
Airmass coefficient 2 (a2)	-0.47 +/- 0.4
Scatter in the residuals of the lightcurve fit is	8.55 %
Best Comparison Star	#1 - [346, 419]
Optimal Aperture	5.25
Optimal Annulus	9.75
Transit Duration (day)	0.089 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.201 ± 0.06 vs 0.14762 (deviation 0.89σ)
Duration fit vs published	0.089 d vs 0.1033 d (deviation 0.6σ)
Mid-time O–C	3.2 min
Depth SNR	3.4
Residual scatter	8.55 %

Light curve

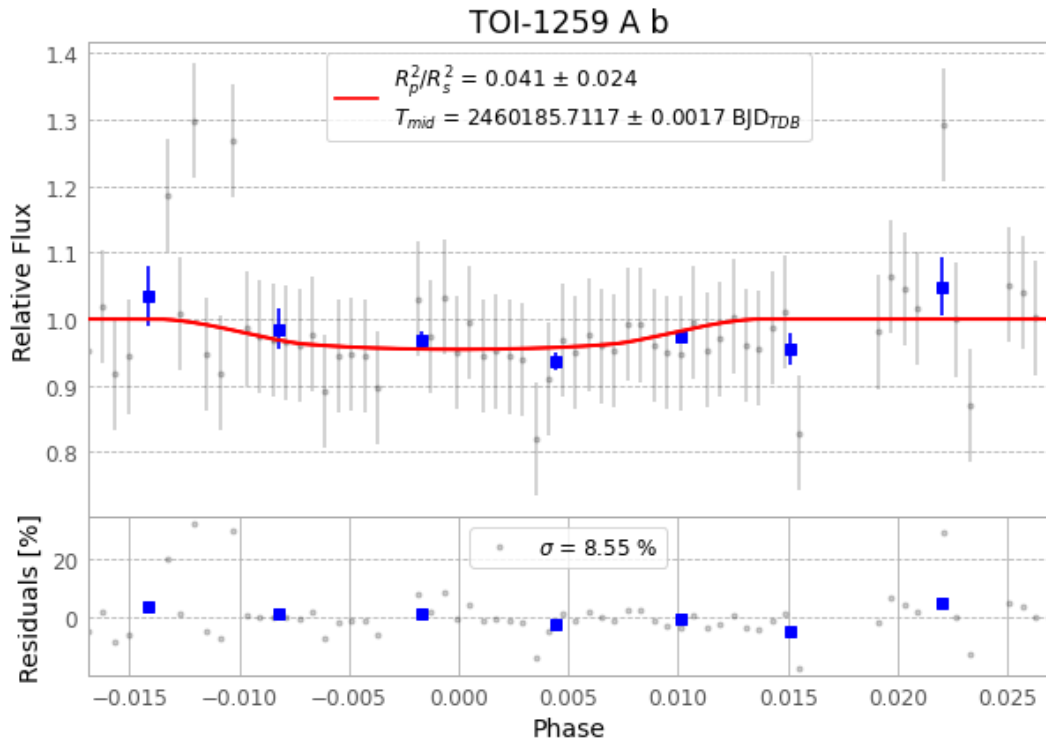


Figure 1 — FinalLightCurve_TOI-1259 A b_2023-08-28.png (SHA-256 daf8cfcb6b3e034d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [261, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d85c3733e9fd8557...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

74 FITS frames (49 MB), TOI1259230829033915.FITS → TOI1259230829071815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1259-230829.log (a5d8ecb3a486...), FinalLightCurve_TOI-1259 A b_2023-08-28.png (daf8cfcb6b3e...), FinalLightCurve_TOI-1259 A b_2023-08-28.csv (aee8e57bb4ab...)

Compute resources

Wall time 232.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:35Z.